AMLET
Reference manual



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Abstract

This document explains how to install and use the software AMLET v0.13.1, designed to estimate multinomial and mixed logit models. The current documentation is unfortunately far from be completed, and is full omissions and possibly errors. Any suggestion concerning this document and/or the software should be send to bastin@iro.umontreal.ca.

The latest version of this document, as well as the software itself, can be retrieved at the internet address http://amlet.slashbin.net.

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Chapter 1

Introduction

1.1 What is AMLET?

AMLET is an acronym for Another Mixed Logit Estimation Tool. As its name suggests, it is a software mainly designed to estimate various kind of mixed logit models. It can be used either as a standalone program, either as a C library. AMLET is an open source project (see Appendix C.1; the last version can be downloaded at the address `http://amlet.slashbin.net`.

1.2 Supported systems

AMLET development is primarily Unix-oriented, while the software can also be used on Windows systems, with some technical limitations[1]. Most of the development and tests have been performed on Linux systems. The software should also runs without major trouble on BSD systems, or Mac OS X. Windows users should be able to compile AMLET using MinGW, available at `http://www.mingw.org`, at the expense of some loss of performance. For Windows users, it is recommended to install Linux in a virtual machine, using for instance VirtualBox, available for free at `http://www.virtualbox.org`. See Section 1.3.3 for more details.

Unless otherwise stated, we will assume in this document that you work on a Unix-like system; the explications are still valid for the other environment. We also assume that you use `bash` as your shell, since it is the default choice for major Linux distributions, as well as `mingw`; some users could have to slighty adapt the explanation if they use csh shells, and its variants. If you do not know what is your

[1] In particular, the timings will be less accurate.

current shell, you probably already use `bash`. It can also be determined by reading the file `/etc/password`.

1.3 Requirements

1.3.1 Hardware requirements

There is no formal minimum requirements for AMLET, since they crucially depend on the size of dataset size and the number of random variables that are present in the model. We however recommand a minimum of 1Go RAM, knowing that the available memory constrains the achievable accuracy. AMLET should work smoothly on any modern computer.

1.3.2 Software requirements

Various software elements are required in order to correctly install AMLET, as well as some optional elements, if you want to install additional tools along with the main package.

C compiler We recommand to use the compiler delivered with your computer; on Linux-machine and mingw, this is typically the GNU C Compiler (gcc).

Compilers The code is mainly written in C and C++, but contains portions written in Fortran, mainly imported from LAPACK. Consequently, both a C and a Fortran compilers are requires; the code has been successfully tested with `gcc` and `gfortran`, available in any Linux distribution.

Unix shell In order to easily perform the installation, autotools (autoconf, automake, libtoolize) are highly recommanded. The detailed installation process can however often be pursued without them. If you are using Mac OS X, you can experience problems with `libtoolize`. Using the development environment proposed in the third-party software collections fink (`http://www.finkproject.org/`) can resolve this issue.

Lex, Yacc Driver parsing with AMLET requires standard lexical analyser and parser. AMLET requires flex, and yacc or bison.

Perl Some optional analysis tools provided with AMLET rely on Perl scripting language.

Oratio, Ophelia These are two companion libraries, that have to be installed first and can be downloaded on `http://amlet.slashbin.net`.

1.3.3 Windows users

The most convenient solution is to install Linux on a Virtual Machine, using for instance VirtualBox (http://www.virtualbox.org). A virtual machine allows to emulate a PC, allowing the installation of a complete operating system. There are of course some overheads in executation time, as it means that one operating system is running inside another one, but a Virtual Machine can be easily uninstalled if no more needed, without affecting the host Operating System. It is therefore the easiest solution.

A more complex solution is to install MinGW, standing for Minimal Gnu system for Windows. MinGW can be download at the address http://www.mingw.org. Pay attention to install all the development packages during the installation, in order to have to required tools to compile the code. It is also a good idea to install a correct text editor. Various editors are available under Windows: Vim, Emacs, Xemacs.

1.4 Acknowledgements

Like for many projects, the development of AMLET has only been possible with the help of many people. First of all, I would like to express my gratitude so Cinzia Cirillo, for the hard work that we have achieved together. This software could not has been written without her advices, based on her huge experience of the practical aspects of the problem. I would also like to thank Philippe L. Toint, who was my Ph.D. advisor during four years, and teached me the most important aspects of nonlinear programming. Finally, my thanks go to Belgian National Fund for Scientific Research, who gave me the grant that made the initial work on this project possible. The development of AMLET has been initiated at the University of Namur (Belgium), continued in the Imperial College London (United Kingdom) and Cerfacs (Toulouse, France). I pursue it during my (very limited) free time at the University of Montreal (Canada).

The documentation of a software is always a difficult exercise. I would therefore like to thank anyone that helped us to improve the present document. In particular, I would like to express my gratitude to Roberta "Speedy" Pellicanò for her suggestions, during my study stay in the University Frederico II of Naples, Italy (June 2004), and Jasper Knockaert. The software has also benefit from many suggestions during its development. The following list is certainly far from be complete; my apologies go to each one I have forgotten. I will cite here (by alphabetical order) Michel Bierlaire and Stéphane Hess.

Finally, the painting reproduced the front cover is due to Pieter van Steenwyck, under the name *Ars longa, vitta brevis*.

1.5 Installation

AMLET is currently distributed in source code only, so that the installation process primarily involves the compilation of the various source files. AMLET first requires the installation of the libraries ORATIO and OPHELIA. We refer to their respective documentations for details about the corresponding installation procedures.

As for the libraries, you can compile AMLET with the commands

```
./configure
make
sudo make install
```

Optimization compilation options can be specified as arguments of `configure`, for instance:

```
./configure CFLAGS=''-O2'' FFLAGS="-O2"
```

Chapter 2

First steps with AMLET

AMLET can be used as a C library or a standalone application. When used as a library, more complex models can be described and more analysis can be performed, but at the expense of some programming efforts. The use as a standalone application is easier, but the range of models that can be estimated is more limited.

2.1 Standalone application

When AMLET is used as a command-line application, it has to be launched from a text-console. The basic usage is

```
amlet [OPTIONS] FILE
```

where `FILE` designs the driver file (see Section 2.2), and `[OPTIONS]` corresponds to optional options, as described in Table 2.1. The options can be used in short or long form; for instance, the command

```
amlet -o log driver
```

has the same effect that

```
amlet --output log driver
```

We briefly describe these options here, and cover them in more details in the relevant sections.

debug is meant for development purposes, delivering additional messages on the standard output in order to help debugging code;

help option makes AMLET displaying some brief help on the screen, and then exits.

short	long	argument	function
`-d`	`--debug`	-	activate debug mode
`-h`	`--help`	-	print help message
`-o`	`--output`	file name	save the log information into a file
`-v`	`--verbose`	-	verbose mode
`-V`	`--version`	-	print version number

Table 2.1: command-line options

output allows the user to define the name of the log file (see Section 5.2).

verbose option informs AMLET to print additional information on the screen during execution.

version prints the version number on the screen, afterwhat the program exits.

2.2 Driver file

The AMLET driver file expresses a specification of the model to calibrate, as well as of the optimization method to apply. The driver has to be written in raw text format; examples of driver files can be found in Appendix A. It basically consists of two kinds of lines: the command ones and the affectations ones. A command line has the following form:

```
command parameters;
```

In other terms if consists of a command followed by the parameters to apply to this command. An affection line has the form

```
variable = value;
```

The value can be an integer, a real, a string or a set, depending on the variable that has to be affected. In some cases, a predefined constant identifier can be used instead of a numerical quantity. An integer variable also accepts a real with an integer value, as `1e3`, and a real variable accepts an integer; a value of another type, or a set given to a variable that expects a single value (and conversely), is an error. A name that is not a variable of AMLET is ignored, with a warning. String values are any sequence of characters, enclosed within double quotes; a correct string would be for example

```
``this is a string''
```

You cannot use double quotes inside a string. We give in Table 2.2 the name of the main variables, in alphabetical order. There will be discussed in more details in the relevant sections.

You can also write comments with the characters # or @. The remaining of the line is then ignored by AMLET, except if these characters are part of a string. The driver file is case unsensitive, except for filenames, depending on the operating system on which AMLET is executed.

Commands

We now briefly review the commands of AMLET. First, the command `output` expresses the name of file where the results will be saved:

```
output ``results'';
```

where `results` has to be replaced with the name that you want. If no `output` command is present in the drive file, the name will be set by default to the name of the driver file followed by `.output`.

The next two commands allow to specify an existing observations file or to generate a file that will be filled with synthetic data. Assuming that your observations file has the name `data`, you can load it into AMLET with the command

```
load ``data'';
```

On the other name, synthetic data can be obtained with the command

```
generate ``data'';
```

The structure of the data files is explained in Section 4.1.

Parameter	Default value	Function
b	$(0,\ldots,0)^T$	starting point
btype		type of the utilities coefficients
bgen		logit parameters for synthetic population generation
bgentype		type of the utilities coefficents during observations generation
censor		0 if all individuals face to all alternatives, 1 otherwise
iddep		dependent variable index
idcensor		variable index corresponding to the alternatives availability
draws	new	new to generate new draws, old to read the draws saved in drawsname by an earlier run; if this file does not exist, new draws are generated and saved in it, and if it does not hold the draws of the model, AMLET stops
drawsname		name of the file where the draws are saved, or read with draws = old
eps		tolerance used for optimization
format	Train96	format of the data file
hessian_update		Hessian approximation method
hessian_approx	false	compute the standard errors with the approximation of the Hessian built by the optimization, instead of the Hessian; not recommended: the approximation depends on the optimization path, and gave standard errors 10 to 40% too small on a panel example
level_accuracy		significance level used for objective accuracy evaluation
method		optimization method
np		number of individuals
nalt		number of alternatives
niter		maximum number of iterations during the optimization process
nobs		total number of observations
nrep		number of draws
nvar		number of variables (coefficients)
problem		problem name
robust		indicates if robust estimation of variance-covariance matrix has to be used
rangen		random numbers generation method
legacy_rng	false	reproduce the random draws of AMLET 0.13.1 (see Section 3.4)
sampling		sampling method
randomization		randomization strategy
seed1		seed of the random generator
validation	false	results validation

Table 2.2: variables

Chapter 3

Discrete choice models

3.1 Supported models

AMLET allows the estimation of multinomial (or conditional) logit models as well as mixed logit models. When use as a standalone program AMLET automatically determines the model type from the utility description. The executable version of AMLET currently supports linear-in-parameters utilities only, i.e., for $i = 1, \ldots, I$, $j = 1, \ldots, J$,

$$U_{ij}(x_{ij}) = \sum_{k=1}^{n} \beta_{ik} x_{ijk} = \beta^T x_{ij},$$

where n is the number of dependant variables. Each component of the vector β can be fixed, constant or random, as summarized in Table 3.1.

`FIXED`	fixed coefficient
`CONSTANT`	constant (non-random) coefficient
`LOGNORMAL`	lognormally-distributed coefficient
`NORMAL`	normally-distributed coefficient
`SPLINE`	spline-distributed coefficient
`TRIANGULAR`	triangularly-distributed coefficient
`UNIFORM`	uniformly-distributed coefficient

Table 3.1: parameters types

3.2 Supported distributions

The following distributions are currently supported for the coefficient of utilities in a mixed logit model and for observations when generating synthetic data.

Lognormal lognormally distributed coefficient; the coefficient is calculated as

$$e^{\mu+\sigma X}$$

where $X \sim N(0,1)$, and μ and σ are the parameters to estimate. Therefore μ and σ are respectively the mean and the standard deviation of the logarithm of the coefficient.

Since lognormally distributed coefficients are always positive; you have to enter the negative of the explanatory variable in your dataset if necessary.

Normal normally distributed coefficient, with the mean and standard deviation being estimated.

Spline *To check.*

Triangular Triangularly distributed coefficient, with the mean and "spread" being estimated. A triangular distribution with mean b and spread a has a zero density outside $[b-a, b+a]$, while it increases linearly from $b-a$ to b and it decreases linearly from b to $b+a$.

Uniform uniformly distributed coefficient, with the mean b and "spread" a being estimated. A uniformly distributed coefficient of mean b and spread a has therefore the distribution of $U[b-a, b+a]$.

You can also use the previous distributions to generate synthetic data, as well as the following one:

Bernouilli an observation can be drawn from a Bernouilli of parameter p ($0 \leq p \leq 1.0$), that has to be specified.

The used distributions are specified in the variables `btype` and `bgentype` as set of identifiers. The truncated normal distributions (`truncated_normal` and `truncated_normal_b`) and the Bernouilli distribution can only be used in `bgentype`, to generate synthetic data.

3.3 Sampling methods

Multi-dimensional integrals are approximated by drawings points from the underlying distributions of the random parameters. Standard Monte-Carlo draws can be used as well as some quasi-Monte Carlo techniques:

- Halton sequences [3],
- Lattice nets [10],
- Modified Latin Hypercube Sampling as proposed in Hess et al. [5],
- Sobol sequences, following the recommendations of Joe and Kuo [6].

These various quasi-Monte Carlo sequences can be generated by using the library ORATIO(). The generation method can be defined by with the variable `sampling`, and is set by default to the standard Monte-Carlo approach. Possible values are summarized in Table 3.2.

Pseudo-random numbers	`monte_carlo`
Halton sequences	`halton`
Modified Latin Hypercube Sampling	`mlhs`
Rank-1 Lattice	`lattice`
Sobol sequences	`sobol`

Table 3.2: sampling methods

The number of random draws to use per individual is defined with the variable `nrep`, and is set by default to 1000. A higher value is sometimes needed, especially if the accuracy and bias on the objective are important. To use more draws, say 10000 per individual, write

```
nrep = 10000;
```

When lattices are used, AMLET uses the tabulated rank-1 lattice with the largest number of points (a prime number) not greater than `nrep`, and sets `nrep` to this number, for instance 887 for `nrep = 1000`.

The methods with a variable sample size (`mcbtrda` and `mcrtrda`) estimate the simulation error as for independent draws. With quasi-Monte Carlo draws, AMLET therefore replaces them by the same methods with a fixed sample size (`mcbtr` and `mcrtr`), with a warning. For the same reason, the correction of the simulation bias (`bias = 1`, the default) is turned off with quasi-Monte Carlo

draws. With Monte Carlo draws, the optimization uses the corrected simulated log-likelihood: the log-likelihood that the results give is the corrected one, and the results also give the simulated log-likelihood without the correction. While Halton are the most popular sequences in discrete choice theory, we recommend the use of Sobol sequences of rank-1 lattices. However, the advantage of quasi-Monte Carlo sequences is not clear, as shown in [10].

3.3.1 Halton sequences

A popular alternative to pseudo-random draws in the field of mixed logit modelling is the use of Halton sequences (Halton [3]). While we strongly recommend the use of Sobol sequences instead of Halton ones, it is possible to select this approach by correspondly setting the variable `sampling`:

```
sampling = halton;
```

One major drawback of Halton sequences is the correlation between sequences associated to successive dimensions when the dimensionality is quite high (order of 10 or superior). Two techniques have been proposed to break such correlations: scrambling (Bhat [1]) and shuffling (Hess et al. [4]). Both of them can be used in AMLET. It is also possible to randomize the sequences by randomly shifting the sequences, for each dimension, as proposed by Bhat [1]. While all these techniques are not strictly speaking randomizations methods, all of them can be set using the variable `randomization`, whose default value is `none`, as summarized in Table 3.3.

`none`	no randomization
`scrambling`	scramble the Halton sequences
`shifting`	randomly shift the Halton sequences
`shifting_scrambling`	randomly shift and scramble the Halton sequences
`shuffling`	randomly shuffle the Halton sequences
`shifting_shuffling`	randomly shift and shuffle the Halton sequences

Table 3.3: Randomization strategies for Halton sequences

In AMLET, all individuals use the same Halton points, each with its own random shift, so that `none` and `shifting` give the same draws. `scrambling` applies the Braaten-Weller permutations (up to dimension 16), and `shuffling` shuffles the order of the points of each individual, dimension by dimension. Up to the 2026 rebuild, AMLET ignored the variable `randomization`; it still does with `legacy_rng = true`.

3.3.2 Sobol sequences

3.3.3 Rank-1 lattices

3.4 Pseudo-random numbers generators

Each method requires the use of a uniform random numbers generator, whose quality is important to ensure quality of the results. Even if a quasi-Monte Carlo is used, the randomization part also involves the use of a pseudo-random numbers generator. The choice of a good one is still a tricky question (see for instance the relevant discussion in L'Ecuyer [7] and McCullough [9]). Unlike many econometric packages, as LIMDEP, AMLET allows the user to choose between several random number generators, defined in the random numbers generator library. Three generators are currently available: the standard minimal one, proposed by Park and Miller [11], the MRG32K3a generator, due to L'Ecuyer, and the generator MT19937 designed by Matsumoto and Nishimura [8]. To date no significant differences have been observed when using one technique instead of another one, while it is known that the sandard minimal generator has severe statistical deficiencies. By default we use the MRG32K3a generator that benefits from strong theoretical properties and is well tested, and our implementation allows to consider substreams. The choice of the random number generator can be defined with the variable `rangen`. Possible values are summarized in Table 3.4.

Standard minimal	`standard`
L'Ecuyer's generator	`ecuyer`
MRG32K3a (default)	`mrg32k3a`
MT19937	`mt19937`

Table 3.4: random number generators

Up to the 2026 rebuild of AMLET 0.13.1, the MRG32K3a generator of ORATIO 0.7.5 differed from L'Ecuyer's: its first component used x_{n-1} instead of x_{n-2}, and the jumps to the next stream or substream were not reduced modulo m. It now produces L'Ecuyer's MRG32K3a sequences, streams and substreams, so that simulated results differ from those of AMLET 0.13.1. To reproduce them, set

```
legacy_rng = true;
```

in the driver file. The legacy generator is not MRG32K3a and its statistical properties are unknown: use it only to reproduce earlier results.

Each random generator produces a sequence of points that mimics draws from an uniform distribution on $(0, 1)$.. These sequences are however deterministic due

to their numerical nature, so they can be reproduced when some initial conditions are fixed. Typically, they can be expressed by a single strictly positive integer, known as the *seed*. This value can be defined in the driver file by means of the variable `seed1`, for example:

```
seed1 = 12345;
```

Most of the time, the seed can be recovered at any point during the random draws generation process, allowing the replication of a sequence starting at any draw of the initial sequence. This is useful for instance when dealing with a newly generated synthetic population, where the same random generator is used for creating the population and the random draws. In this case, the output file will contain both the initial seed and the seed after the population generation. Note however that the MT19937 cannot be restarted with a single seed, except the initial one.

The default value for `seed1` is 12345. If a seed value of 0 is used, AMLET will determine a seed based on the computer clock, that will be used to initiate the pseudo-random draws sequence.

For more information, we refer the reader to the documentation accompying the ORATIO library.

3.5 Results validation

The source file `validation.c`, associated to the header file `validation.h`, contains basic validation routines concerned with validation of the estimation results, on a statistical point of view. The validation is currently quite simple, as it simply repeats `nsim` times the simulation of the objective function, using a different set of random realizations for each simulation. The mean, standard deviation, and a confidence interval at 0.9 for the objective function are given on output. In order to activate the validation function, you need to set the variable `validation` to `true` in the driver file:

```
validation = true;
```

Chapter 4

Data files

4.1 Data files

Every model needs data in order to proceed to its calibration, so the possibilty to enter data in a convenient format is important. AMLET unfortunately currently supports basic formats only, even if development is planned on this issue. The supported formats reflect the original purpose of the package, which was only testing a new optimisation method.

4.1.1 GAUSS-like formats

Preliminary AMLET design mimics Train's GAUSS codes for mixed logit estimation (cross-sectional and panel versions). These routines are available at the address `http://elsa.berkeley.edu/~train/software.html`. We also refer the reader to Revelt and Train [12], and Train [13], for more details. AMLET has currently no capabilities to construct new data from a preexisting dataset, for instance dummy observations for alternatives specific constants, so please be sure that your data are completely specified before using AMLET.

The format of the data file is specified by assign to the variable `format` of the driver file the value `Train96` of `Train99`, the number being a reference to Train's code release year. For instance, the following line

```
format = Train96;
```

indicated that data are stored in `Train96` format, which is also taken as default if no format is specified in the driver file.

```
   0    1    0    0
   0    0    1    0
   0    0    0    1
 1.6 -2.0  3.2  4.1
   0  2.5  1.0  0.8
-1.2 -1.5 -0.4 -1.6
   0    1    0    0
   0    1    0    0
   0    0    1    0
   0    0    0    1
 2.4 -3.0  2.3  3.0
   0  1.7  1.2  1.5
-1.6 -2.0 -1.2 -2.4
   0    0    0    1
```

Figure 4.1: Train96 Format

Train96 format

This format is primarily designed for cross-sectional data, as illustrated in Figure 4.1.1. In this figure, each indiviual is facing four alternatives, corresponding four columns. The six first rows of each record (containing seven rows) corresponds to the independent variables. Since each individual is assumed to deliver one observation, this format cannot be used for panel data. An additional row is devoted to the dependent (decision) variable, with one on the column corresponding to the chosen alternatives, and 0 elsewhere. The line position in each record must be indicated with the variable `iddep` in the driver file. For Figure 4.1.1, we would have

```
iddep = 7;
```

By default, all alternatives are available to each individual. If you want to specify alternative availabilities, you have to set the censor variable to 1 in the driver file:

```
censor = 1;
```

The variable `idcensor` then represents the line describing alternative availabilities. The corresponding line in the driver file is simply a 0-1 line, with one on available alternatives, and 0 elsewhere.

Train99 format

This format is used for panel data. It adds to the base name the suffixes `_x`, `_y` and `_times`.

Chapter 5

Output files

The estimation process would be useless if no exploitable results were delivered to the user. The software therefore delivers various files at the of the optimisation process. Their content if meaning is described below

5.1 Results file

5.1.1 Definition

The result of the estimation process is written in the file defined by the command `output` inside the driver file (see Section 2.2). If no name is precised, the default name is the name of the driver file followed by `.output`.

The main characteristics of the problem are first summarized. It is always a good idea to name the problem under consideration, by setting the variable `problem` in the driver file, for instance

```
Problem = ``test'';
```

5.1.2 Content

t-statistics

The t-statistics can be computed in various ways, depending on the options present in the driver file. Each method involves the computation of the information matrix. However, due to the sampling errors, robust t-statistics are not necessarily better than those obtained with the outer approximation. The sampling errors are indeed amplified when computing second-order derivatives.

5.2 Log file

AMLET generates a log file during its execution. It will contain detailed information on the process, from initialisation tasks to a complete recording of the optimisation iterates. The default name of the log file is `amlet.log`, but this name can be change with the `-o` and `--option` options at the start of the software (see Chapter 2).

Appendix A

Example of driver file

We illustrate the behaviour of AMLET on a 1987 dataset made of 210 non-business trips between Sydney, Canberra and Melbourne. In this dataset, each traveller chooses a mode from four available alternatives (plane, car, bus and train). This dataset is described in more details in Greene [2] (Example 19.18), and is available at the address
http://www.ats.ucla/stat/limdep/how_to_run_statistical_analysis_.htm.
A formatted version for AMLET is present inside the directory `testing` of the AMLET source code, as the file `model_australia`.

A.1 Multinomial logit

A.1.1 Driver file

```
# Problem name;
PROBLEM = ``Australia problem'';

# name of the output file
output "results_mnl";

# number of observations
nobs = 210;
np = 210;

# number of aternatives
nalt = 4;

# number of variables
```

```
nvar = 7;

# number of variable that is the dependant variable (choice)
iddep = 7;

# 1 if all people do not face nalt alternatives, else 0
censor = 0;

# maximum number of iterations in the maximization
niter = 150;

# tolerance for convergence
eps = 1e-6;

# type of parameters
btype = { CONSTANT, CONSTANT, CONSTANT,
          CONSTANT, CONSTANT, CONSTANT };
# starting point
b = { 0, 0, 0, 0, 0, 0 };

# data file
load "model_australia";

method = MCBTR;
hessian_update = BFGS;

iapprox = false;
robust = false;
```

A.2 Mixed logit

A.2.1 Driver file

```
# Problem name;
PROBLEM = "Australia problem";

# name of the output file
output "results_mmnl";

# number of observations
nobs = 210;
np = 210;

# number of aternatives
```

```
nalt = 4;

# number of variables
nvar = 7;

# number of variable that is the dependant variable (choice)
iddep = 7;

# 1 if all people do not face nalt alternatives, else 0
censor = 0;

# maximum number of iterations in the maximization
niter = 150;

# tolerance for convergence
eps = 1e-6;

# type of parameters
btype = { CONSTANT, CONSTANT, CONSTANT,
          CONSTANT, NORMAL, CONSTANT };
# starting point
b = { 0, 0, 0, 0, 0, 0, 0 };

# data file
load "model_australia";

method = MCBTRDA;
hessian_update = BFGS;

iapprox = false;
robust = true;
```

Appendix B

Numerical issues

B.1 Choice probability computation

We start from the logit expression

$$P_{ij} = \frac{e^{U_\ell}}{\sum_{k \in \mathcal{A}_i} e^{U_k}}. \tag{B.1}$$

If some of the utilities are too large, the computation of (B.1) can face some overflow. A simple way to avoid this problem is to substract some quantity α since

$$\frac{e^{U_\ell - \alpha}}{\sum_{k \in \mathcal{A}_i} e^{U_k - \alpha}} = \frac{e^{U_\ell}}{\sum_{k \in \mathcal{A}_i} e^{U_k}}.$$

Moreover, is all utilities are too small, they could be evaluated to 0, leading to the ratio $0/0$ in (B.1). A simple way to circumvent this issue is to normalize with respect to the highest utility, i.e. take

$$\alpha = \max\{U_k,\ k \in \mathcal{A}_i\} - \beta.$$

We want to keep each term e^{U_k}, $k \in \mathcal{A}_i$, in the same range, so β has to be kept low. Since $de^x/dx = e^x$, the precise value of β does not significantly affect small utilities, so a simple, while practical, choice is to set β to 0. As a consequence, $e^{U_k - \alpha} \in (0, 1]$ for all $k \in \mathcal{A}_i$.

B.2 Data file problems

Appendix C

Legal aspects

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